

MONTREAL-EST, QC, Canada

WMO: 716751

Lat: 45.630N

Lon: 73.550W

Elev: 165

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -4.4	(d) -22.0	(e) 1.6	(f) -9.6	(g) -17.1	(h) 2.2	(i) -4.8	(j) 22.8	(k) 30.8	(l) 20.6	(m) 27.0	(n) 8.5	(o) 260	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	86.9	69.8	84.2	68.1	81.7	66.9	72.9	81.9	71.1	79.1	69.6	77.2	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	111.4	76.6	68.4	104.9	75.7	66.8	99.1	74.7	36.6	82.3	35.1	78.8	33.7	76.9	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.3	(b) 17.0	(c) 15.2	(e) -13.2	(f) 90.6	(g) 5.5	(h) 2.6	(i) -17.1	(j) 92.5	(k) -20.3	(l) 94.0	(m) -23.4	(n) 95.5	(o) -27.4	(p) 97.3		
(4) 19.3	(5) 17.0	(6) 15.2	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	16.1	18.9	29.2	43.2	56.8	67.0	70.5	69.3	61.5	48.6	36.3	24.0		
	DBStd	21.07	12.58	10.64	10.34	8.39	7.56	7.11	5.08	5.59	7.13	8.10	8.63	10.32		
	HDD50	4177	1050	871	646	233	20	0	0	0	3	127	419	808		
	HDD65	7729	1515	1291	1110	654	273	62	10	22	149	510	861	1272		
	CDD50	2442	0	0	1	30	231	509	635	597	348	84	7	0		
	CDD65	521	0	0	0	1	19	121	180	154	43	3	0	0		
	CDH74	3702	0	0	0	11	172	963	1229	1049	267	11	0	0		
	CDH80	904	0	0	0	1	35	271	285	250	62	0	0	0		
(14) Wind	WSAvg	7.5	8.1	7.9	8.1	8.1	7.8	7.2	6.9	6.4	6.8	7.5	7.7	7.8		
(15) Precipitation	PrecAvg	40.5	2.9	2.4	2.6	3.4	3.5	3.7	3.8	3.7	3.7	4.0	3.2	3.6		
	PrecMax	51.3	5.8	4.3	4.8	6.2	6.8	6.4	6.0	7.7	7.5	7.4	5.3	6.1		
	PrecMin	31.9	1.1	1.0	0.8	0.7	1.7	1.6	2.1	1.5	1.1	1.3	0.6	1.1		
	PrecStd	4.8	1.3	0.8	0.9	1.5	1.3	1.1	0.9	1.4	1.5	1.6	1.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	44.3	58.1	74.7	84.8	89.8	89.1	89.5	87.0	75.3	62.9	52.0		
		MCWB	40.6	40.1	48.8	59.2	63.9	70.0	73.9	70.9	68.4	64.4	58.2	49.4		
	2%	DB	42.6	40.6	51.0	67.0	78.7	86.4	85.7	85.5	80.6	70.0	57.8	44.5		
		MCWB	38.8	37.1	43.0	52.0	58.9	68.1	70.9	69.2	67.1	59.7	53.3	41.5		
	5%	DB	38.0	37.5	46.0	61.2	74.2	83.1	83.2	82.6	76.4	65.5	52.6	40.4		
		MCWB	35.1	34.9	40.0	48.5	57.8	66.9	68.4	67.6	64.9	57.2	48.6	38.0		
	10%	DB	33.8	34.6	42.7	56.2	70.1	79.8	80.6	79.8	72.9	61.0	47.9	37.2		
		MCWB	32.3	32.1	38.1	45.7	56.6	64.8	67.2	66.6	63.2	54.2	43.7	35.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	42.3	50.9	61.3	67.5	73.7	76.2	74.2	71.3	65.6	58.4	50.1		
		MCDB	46.2	43.9	55.4	68.1	76.9	82.4	85.6	85.2	78.1	73.0	61.8	52.8		
	2%	WB	36.7	38.4	44.6	55.1	64.2	70.9	73.8	72.2	69.0	61.1	53.9	42.2		
		MCDB	39.4	40.7	48.5	63.4	72.5	79.5	82.5	81.5	77.4	67.4	57.3	44.6		
	5%	WB	34.0	35.5	41.2	50.3	61.7	69.1	71.7	70.5	66.8	57.5	49.0	38.3		
		MCDB	35.6	37.6	45.6	58.3	70.6	77.4	79.4	78.3	74.5	64.4	52.7	40.4		
	10%	WB	30.1	32.4	38.4	46.6	58.9	67.2	69.8	68.8	64.6	54.4	44.2	35.3		
		MCDB	32.4	34.8	42.2	53.9	67.2	75.6	77.1	76.5	71.1	59.7	47.4	37.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	14.4	13.8	15.9	17.0	16.6	15.3	15.9	15.8	13.1	10.8	11.2		
		MCDBR	17.0	14.9	18.0	24.1	22.8	20.6	18.0	18.5	19.3	18.8	17.2	13.1		
		MCWBR	19.5	14.0	13.2	14.2	10.3	9.1	8.2	8.1	9.6	11.4	14.4	12.5		
	5% WB	MCDBR	16.9	15.4	18.2	21.2	18.9	16.5	15.3	16.4	16.3	18.3	16.6	13.1		
		MCWBR	16.6	14.7	14.6	14.7	11.3	9.8	8.8	9.4	9.9	12.6	14.8	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.289	0.334	0.363	0.390	0.439	0.428	0.414	0.374	0.336	0.307	0.293		
	taud		2.349	2.250	2.263	2.353	2.339	2.206	2.244	2.280	2.380	2.492	2.487	2.299		
	Ebn at Noon		274	282	283	284	279	264	266	265	267	264	253	241		
	Edh at Noon		23	32	37	37	39	44	42	39	33	25	21	23		
(44) All-Sky Solar Radiation	RadAvg		457	756	1152	1390	1624	1786	1790	1581	1250	722	436	316		
	RadStd		53	80	95	159	155	134	136	74	103	63	56	27		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page